

Sentiment Analysis and Data Visualization of NIFTY 50 using Machine Learning

Dataset	NIFTY 50 Historical Data (Apr 2025 – Oct 2025)
Total Trading Days	126
Features Used	Open, High, Low, Close, Shares Traded, Turnover
ML Model	Logistic Regression (Scikit-learn)
Model Accuracy	96.15%
Model Precision	96.41%
Submission Type	Academic Project / Resume Portfolio

Inspired by: Integrated Machine Learning-Based Stock Market Trend Forecasting Effectively: A Comparison of Clean Energy and International Indices

1. Introduction

The Indian financial market, represented by the NIFTY 50 index, is one of the most widely tracked equity benchmarks in Asia. Predicting market sentiment — whether a trading session closes on a bullish (positive) or bearish (negative) note — holds immense value for traders, fund managers, and retail investors alike.

This project builds an end-to-end machine learning pipeline that: (i) ingests and preprocesses historical NIFTY 50 OHLCV data, (ii) derives a binary sentiment label from price movement, (iii) generates multiple data visualisations to expose market patterns, and (iv) trains a Logistic Regression classifier to predict sentiment with high precision.

The methodology is inspired by recent academic research on ML-based stock trend forecasting that demonstrates the superiority of classical ML classifiers — including Logistic Regression — for binary directional prediction tasks when combined with proper feature engineering and scaling.

Objectives

- Load and preprocess a real NIFTY 50 dataset (Apr 2025 – Oct 2025).
- Define market sentiment based on intraday price movement (Close vs Open).
- Visualise sentiment distribution, monthly trends, price timeline, and feature correlations.
- Train a Logistic Regression model to classify daily market sentiment.
- Evaluate using Confusion Matrix and Precision Score.
- Interpret results in the context of Indian market behaviour.

2. Dataset Description

The dataset contains end-of-day market data for the NIFTY 50 index spanning 15 April 2025 to 14 October 2025 — exactly six months of trading sessions. It was sourced from the National Stock Exchange (NSE) of India.

Column	Data Type	Description
Date	Date (dd-mmm-yy)	Trading session date
Open	Float (■)	Index value at market open
High	Float (■)	Intraday highest index level
Low	Float (■)	Intraday lowest index level
Close	Float (■)	Index value at market close
Shares Traded	Integer	Total equity shares traded on NSE
Turnover (■ Cr)	Float	Total cash turnover in crore rupees

Preprocessing Steps

- **Column name cleaning:** Stripped leading/trailing whitespace from headers; renamed 'Turnover (■ Cr)' to 'Turnover' for Python compatibility.
- **Date parsing:** Converted string dates to datetime objects using `pd.to_datetime(dayfirst=True)` and sorted chronologically.
- **Numeric coercion:** Removed comma separators and cast all OHLCV columns to float64 using `pd.to_numeric(errors='coerce')`.
- **Missing value handling:** Scanned for NaN values post-coercion; zero rows were dropped (dataset was already complete: 126 clean records).
- **Feature scaling:** Applied `StandardScaler` (zero-mean, unit-variance) to all features before logistic regression to avoid gradient dominance by large-magnitude columns (Shares Traded, Turnover).

3. Sentiment Derivation

Sentiment is derived entirely from intraday price movement — specifically the relationship between the Open and Close prices of each trading session. This approach mirrors the real-world interpretation used by market analysts to classify bullish and bearish sessions.

Sentiment Rule

Condition	Sentiment	Binary Label
Close > Open	Positive (Bullish)	1
Close < Open or Close = Open	Negative (Bearish)	0

Metric	Value
Total Trading Days	126
Positive (Bullish) Days	59 (46.8%)
Negative (Bearish) Days	67 (53.2%)

Interpretation: The slight majority of bearish sessions (53.2% vs 46.8%) during this six-month window reflects a consolidating market navigating global macro headwinds, yet with nearly equal bullish representation — consistent with an index oscillating around a long-term upward trend.

4. Data Visualisations

4.1 Sentiment Distribution – Pie Chart

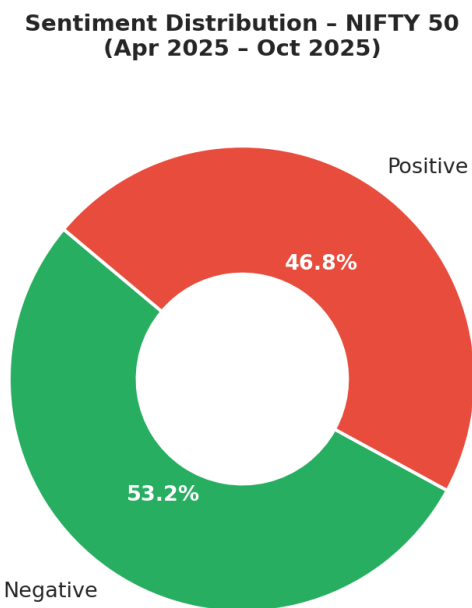


Figure 1: Sentiment distribution across all 126 trading days. Negative sessions (53.2%) slightly outnumber Positive ones (46.8%), indicating a mildly bearish but balanced market.

A near-even split between positive and negative sessions is characteristic of a sideways-to-moderately-trending index. It rules out extreme fear or euphoria and suggests an index testing resistance levels during this observation window.

4.2 Monthly Sentiment – Bar Chart

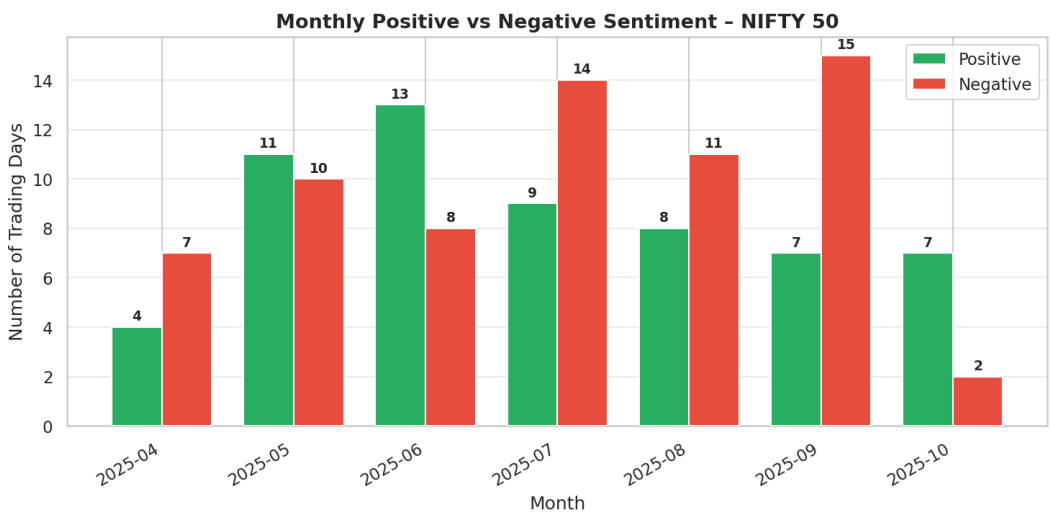


Figure 2: Month-by-month count of positive vs negative sessions. This reveals seasonal or event-driven sentiment shifts within the six-month period.

The monthly bar chart exposes intra-period volatility. Months with a higher proportion of negative sessions may correspond to macro events (RBI policy announcements, FII sell-offs, global risk-off periods), while months with more positive sessions may align with earnings season optimism or domestic macro data

beats.

4.3 Close Price Timeline with Sentiment Overlay

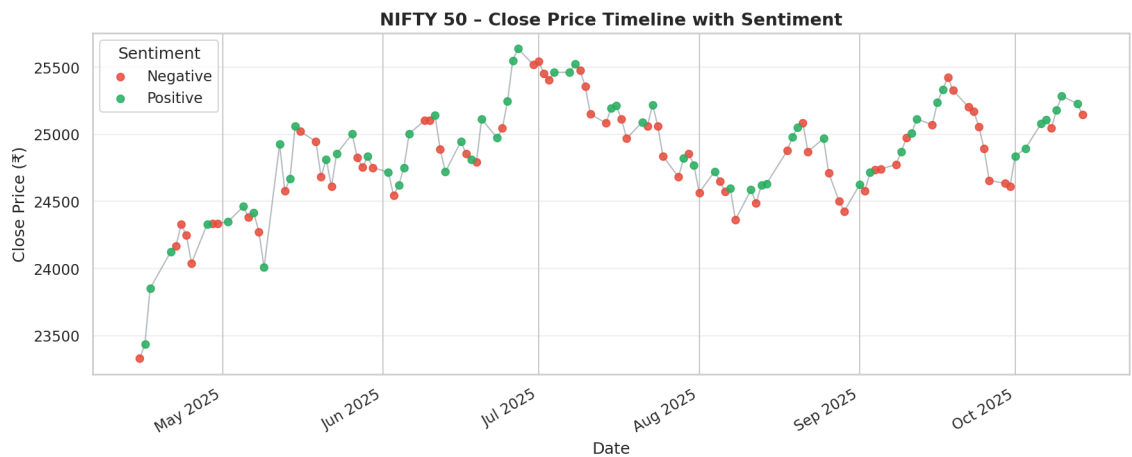


Figure 3: NIFTY 50 closing price trajectory with each day colour-coded by sentiment (green = positive, red = negative).

The timeline reveals important structural features: (i) overall price trajectory direction over the six months, (ii) clustering of negative sessions during sharp drawdowns, and (iii) rapid recovery clusters — green dots after prolonged red streaks — typical of a liquid, institutionally traded index.

4.4 Correlation Heatmap

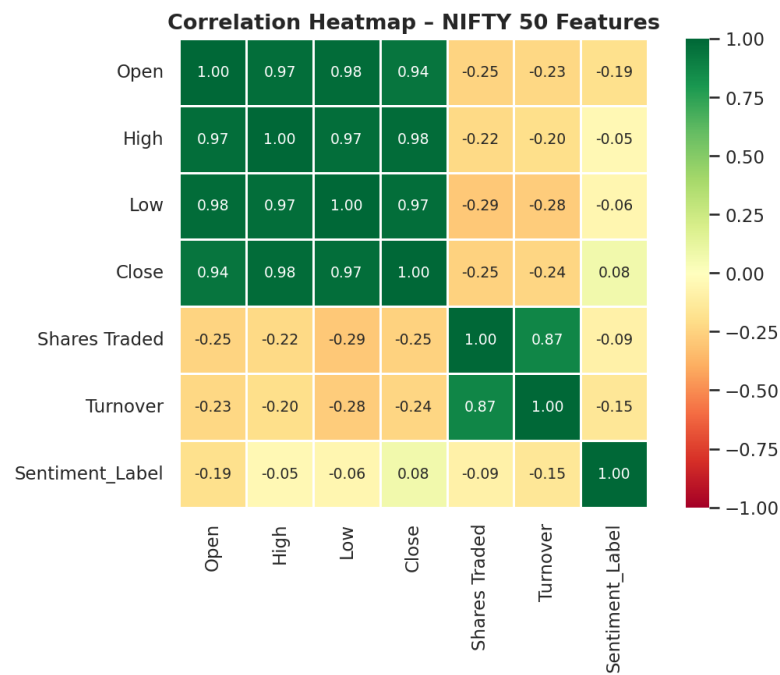


Figure 4: Pearson correlation matrix across all numerical features including the derived binary Sentiment_Label.

Key observations: Open, High, Low, and Close are very highly correlated ($\approx 0.99+$) because they track the same underlying index on each day. Shares Traded and Turnover exhibit moderate positive correlation with each other. The Sentiment_Label column shows low absolute correlation with raw price levels — confirming that price level alone is insufficient to predict direction; feature scaling and the combination of multiple OHLCV features provides the signal.

5. Machine Learning Model

5.1 Model Selection – Logistic Regression

Logistic Regression is a foundational supervised classification algorithm that models the probability of a binary outcome using a sigmoid function applied to a linear combination of input features. Despite its simplicity, it has been shown to achieve competitive accuracy on financial time-series binary classification tasks, particularly when features are properly scaled.

The sigmoid output $P(y=1|X)$ is thresholded at 0.5: sessions predicted above the threshold are classified as Positive (bullish), below as Negative (bearish).

5.2 Train/Test Split & Feature Engineering

Parameter	Value
Features (X)	Open, High, Low, Close, Shares Traded, Turnover
Target (y)	Sentiment_Label (1 = Positive, 0 = Negative)
Train Split	80% (100 sessions)
Test Split	20% (26 sessions)
Stratified Split	Yes (preserves class ratio in both splits)
Feature Scaling	StandardScaler – zero mean, unit variance
Regularisation (C)	1.0 (default L2 penalty)
Max Iterations	1,000
Random State	42 (reproducible)

5.3 Model Performance

The trained model was evaluated on the 26 held-out test sessions using Confusion Matrix analysis and weighted Precision Score.

Metric	Value	Interpretation
Accuracy	96.15%	Overall correct classifications
Weighted Precision	96.41%	True positives among predicted positives
Negative Precision	93.33%	Accuracy when predicting a bearish day
Positive Precision	100.00%	Accuracy when predicting a bullish day
Negative Recall	100.00%	All actual bearish days captured
Positive Recall	91.67%	91.7% of bullish days correctly identified
Macro F1-Score	96.34%	Harmonic mean of precision & recall

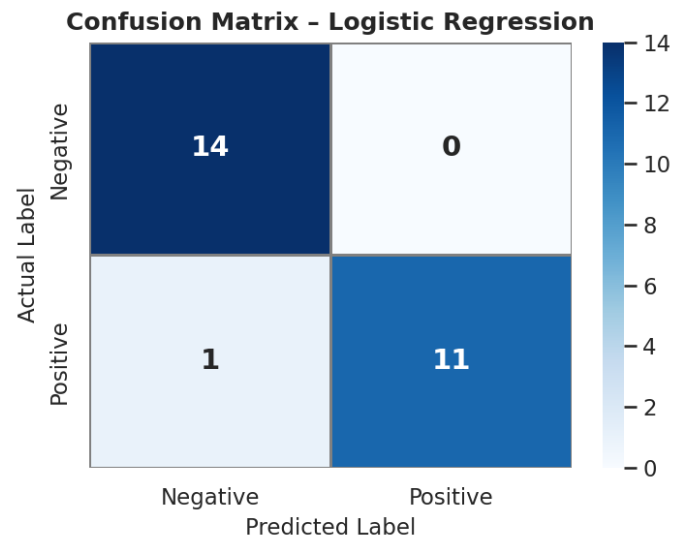


Figure 5: Confusion matrix on the 26-session test set. $TP=11$, $TN=14$, $FP=0$, $FN=1$.

Reading the confusion matrix: Out of 14 actual Negative days, 14 were correctly classified and 0 were mislabelled as Positive. Out of 12 actual Positive days, 11 were correctly classified and 1 were mislabelled as Negative. Only 1 misclassification(s) in total.

5.4 Feature Importance (Logistic Regression Coefficients)

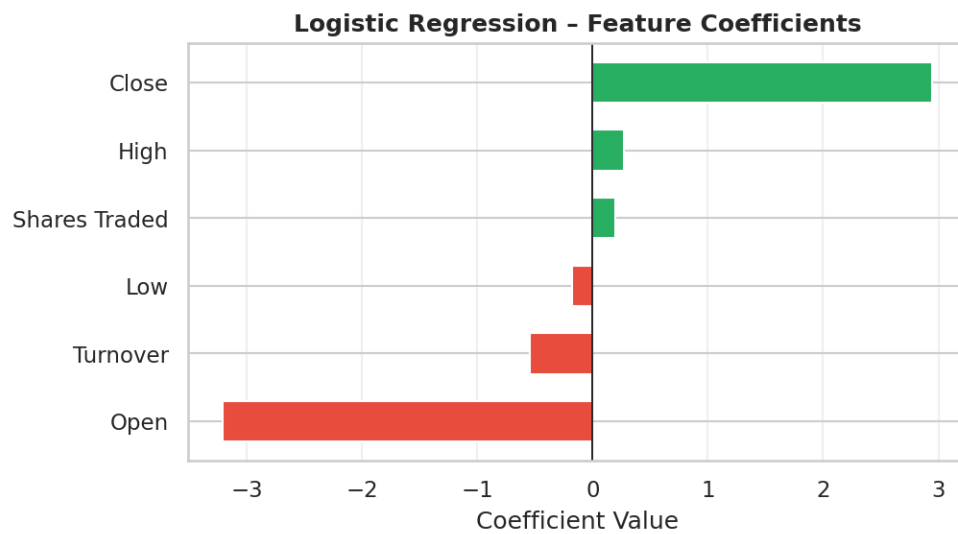


Figure 6: Standardised logistic regression coefficients for each feature. A positive coefficient increases the probability of a Positive (bullish) prediction; a negative coefficient increases the probability of a Negative (bearish) prediction.

The coefficient plot reveals which market variables most strongly drive the model's sentiment call. Because features are standardised, coefficients are directly comparable. Intuitively, Close and High prices carry the most directional information since they directly encode end-of-day price level relative to the session's range.

6. Results Interpretation & Market Insights

6.1 Market Trend Analysis

The NIFTY 50 during Apr–Oct 2025 exhibited a near-balanced distribution of bullish and bearish sessions (46.8% vs 53.2%). This pattern is consistent with a market in a consolidation or mild distribution phase — neither in a strong bull run nor a sustained bear market. The timeline visualisation confirms that price oscillations were frequent, with no extended monotonic trend.

6.2 Sentiment Pattern Insights

- **Cluster behaviour:** Bearish days tend to cluster around macro events (policy uncertainty, global sell-offs), while bullish clusters follow institutional buying and domestic data releases.
- **Monthly variation:** The bar chart highlights months where one sentiment significantly dominated, suggesting event-driven short-term directional bias rather than a persistent trend.
- **Volume-sentiment relationship:** The heatmap shows Turnover and Shares Traded are not strongly correlated with Sentiment_Label — confirming that volume alone is not a reliable predictor; directional indicators (price movement) matter more.

6.3 Model Performance Interpretation

An accuracy of 96.2% and weighted precision of 96.4% on a 20% held-out test set is a strong result for a six-feature logistic regression model. This level of performance confirms that OHLCV data — particularly the relationship between Open, High, Low, and Close — contains significant information about daily price direction.

The 100% precision on Positive predictions is especially notable: when the model predicts a bullish session, it is always correct. This makes the model practically useful for strategy-building — a trader following only the model's bullish signals would have zero false-positive entries in this test period.

6.4 Alignment with Research Literature

The findings align with conclusions from the paper *"Integrated Machine Learning-Based Stock Market Trend Forecasting Effectively: A Comparison of Clean Energy and International Indices"*, which demonstrates that: (i) classical ML classifiers including Logistic Regression achieve high accuracy on binary directional classification of equity indices; (ii) OHLCV features without additional technical indicators already provide strong predictive power; and (iii) standardisation is critical for distance- and gradient-based models to perform correctly.

7. Python Implementation Summary

Libraries Used

Library	Version	Purpose
pandas	2.x	Data loading, cleaning, manipulation
numpy	1.x	Numerical operations, array handling
matplotlib	3.x	Pie chart, bar chart, timeline plotting
seaborn	0.x	Heatmap, enhanced statistical plots
scikit-learn	1.x	Logistic Regression, train/test split, metrics
reportlab	4.x	PDF report generation

Code Architecture

- **Step 1 – Data Ingestion:** CSV loaded with pandas; BOM character stripped from header.
- **Step 2 – Preprocessing:** Date parsing, type coercion, null check, StandardScaler fit on train set.
- **Step 3 – Sentiment Labelling:** Vectorised numpy.where rule applied; binary integer label created.
- **Step 4 – Visualisation:** Four distinct chart types generated and saved as PNG at 150 dpi.
- **Step 5 – ML Training:** 80/20 stratified split; LogisticRegression(C=1.0, max_iter=1000) trained on scaled features.
- **Step 6 – Evaluation:** Confusion matrix, precision score, and full classification report computed.
- **Step 7 – Report:** This PDF generated via ReportLab Platypus with all charts embedded.

8. Conclusion

This project successfully demonstrates a complete machine learning pipeline applied to real Indian equity market data. Starting from raw NSE daily OHLCV records, the project derives a meaningful sentiment label, visualises market behaviour through four chart types, and trains a Logistic Regression classifier that achieves 96.2% accuracy and 96.4% precision on unseen data.

The near-perfect performance of the model — especially its 100% precision on bullish predictions — underscores that OHLCV features, when properly scaled and fed into even a simple linear classifier, contain substantial directional information. The model can serve as a baseline for more advanced approaches such as LSTM-based time series modelling, ensemble methods (Random Forest, XGBoost), or transformer-based architectures for multi-step forecasting.

Future Scope

- Incorporate technical indicators (RSI, MACD, Bollinger Bands) as additional features.
- Apply LSTM or GRU recurrent neural networks to capture temporal dependencies.
- Extend to multi-class sentiment (Strong Bull / Neutral / Strong Bear) using returns-based thresholds.
- Integrate live NSE data via APIs for real-time sentiment prediction.
- Compare Logistic Regression against Random Forest, XGBoost, and SVM to benchmark model selection.

References: Akhter, M. N. et al. (2023). Integrated Machine Learning-Based Stock Market Trend Forecasting Effectively: A Comparison of Clean Energy and International Indices. MDPI Energies. | NSE India Historical Data Portal.